# EXPERIMENT 3

## Build a Vector Database using FAISS and Perform Similarity-based Retrieval

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## AIM

To create a FAISS vector index and retrieve documents whose vectors are closest to a query vector.

## PROCEDURE

1. Install and import FAISS and NumPy.
2. Create sample documents and illustrative numerical vectors.
3. Build a FAISS index using the vector dimension.
4. Add the document vectors to the index.
5. Search the index using a query vector.
6. Display the nearest documents and their distances.

## PROGRAM

```
# Install first: pip install faiss-cpu numpy
import faiss
import numpy as np

documents = [
    "Python programming",
    "Machine learning",
    "Database management"
]

# Illustrative numerical embeddings
vectors = np.array([
    [1.0, 0.0, 0.0],
    [0.0, 1.0, 0.0],
    [0.0, 0.0, 1.0]
], dtype="float32")

index = faiss.IndexFlatL2(3)
index.add(vectors)

query_vector = np.array([[0.0, 0.9, 0.1]], dtype="float32")
distances, indices = index.search(query_vector, k=2)

print("Stored Documents:", documents)
print("\nTop Similar Documents:")
for position in range(len(indices[0])):
    doc_id = indices[0][position]
    print(documents[doc_id],
          "Distance:", round(float(distances[0][position]), 3))
```

## OUTPUT

```
query_vector = np.array(query_vector, dtype='float32')

distances, indices = index.search(query_vector, 2)

print("Query:", query)
print("\nSimilar Documents:")

for i in indices[0]:
    print(documents[i])
```

```
Collecting faiss-cpu
  Downloading faiss_cpu-1.15.1-cp310-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.w
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: numpy>=1.25 in /usr/local/lib/python3.13/dist-packages (
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dist-packages (fr
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.13/dist-p
Requirement already satisfied: cloudpickle>=3.0 in /usr/local/lib/python3.13/dist-packa
Downloading faiss_cpu-1.15.1-cp310-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 18.8/18.8 MB 68.1 MB/s  0:00:00
Installing collected packages: faiss-cpu
Successfully installed faiss-cpu-1.15.1
Query: Artificial intelligence and machine learning

Similar Documents:
Machine learning is used in AI
The earth revolves around the sun
```

## RESULT

Thus, a FAISS vector index was created and the nearest documents were retrieved using vector similarity.